CAPSTONE · AI FOR GOOD · WEEK 5

The grid is the bottleneck, *not the turbines*

Europe can build wind farms faster than it can build the wires to carry their power. That gap decides how much of your 2040 is physically possible.

Dr. Jan Erik Meidell · Wednesday 4 November 2026 · online · 18:15–20:00 · keep the shared canvas open on a second screen

TONIGHT

One hour forty-five, two exercises

18:18	A vote: turbines or wires?
18:21	Two clocks, and the size of the queue
18:32	What congestion costs: redispatch, curtailment, constraint payments
18:45	Negative prices and the capture rate
18:55	Interconnection, and a blackout
19:02	Exercise 1 · where does each of your worlds hit the grid?
19:15	Break · 5 minutes
19:20	The grid's own supply chain · offshore grids
19:30	The balancing loop, and what breaks it
19:40	Exercise 2 · one grid indicator per world
19:52	Into Phase 1, due Sunday · close

VOTE IN THE CHAT · T OR W

What slows European wind more today?

T: building enough turbines. W: building enough wires to connect them and carry their power.

Tonight's answer: **the wires**. The International Energy Agency counts at least **3,000 GW** of renewable projects waiting for a grid connection worldwide, 1,500 GW of them already advanced. That is about five times the wind and solar the world added in 2022.

IEA, *Electricity Grids and Secure Energy Transitions*, October 2023

TWO CLOCKS

A wind farm is quicker to build than the line it needs

Renewable project

1–5 years

to plan, permit and build

Transmission line

5–15 years

to plan, permit and build

If the two clocks are not started together, the farm is finished first and waits.

IEA, 2023. EU law now caps permits for renewables at 12 months onshore and 2 years offshore inside acceleration areas (RED III, Directive 2023/2413). There is no such cap that makes a 700 km line quick.

AN ØRSTED EXAMPLE

Built, and waiting for the cable

Borkum Riffgrund 3, 913 MW, Germany's largest offshore wind farm: the turbines were up, but the grid operator's export link (TenneT's DolWin5) was not. Commercial operation slipped to early 2026.

Who builds what

In Germany and the Netherlands the transmission system operator builds the offshore connection; the developer builds the farm. Two companies, two timetables, one revenue stream.

What the developer loses

Every month of delay is a month of capital spent and no power sold. Week 4: the cost of capital is the biggest line in an offshore wind farm's costs.

As reported in the trade press, 2025. Check the current status in Ørsted's latest interim report before you cite it.

THE SIZE OF THE JOB

Europe's grid bill to 2030

€584 bn

of grid investment needed in the EU this decade, most of it in local distribution grids (€375–425 bn)

400 → 1,000 GW

wind and solar in the EU, 2022 → at least 2030: two and a half times as much, in eight years

~40%

of distribution grids are more than 40 years old

×2

cross-border capacity is to double by 2030

European Commission, EU Grid Action Plan, COM(2023)757, 28 November 2023

THE QUEUE

Projects waiting for a connection

WHERE	WHAT	SOURCE
Great Britain	Over **738 GW** in the queue before reform. "First come, first served" became **"first ready, first connected"** (approved April 2025): projects must show land rights and planning, and fit the 2030 plan. The reformed queue: **381.5 GW**, about half.	NESO
EU, 10 countries that publish data	Almost **700 GW** queued; at least **120 GW** of planned renewables at risk of missing 2030	Ember, April 2026

A queue is not a pipeline. Many projects in it will never be built, and good ones wait behind speculative ones.

GERMANY · WIND IN THE NORTH, INDUSTRY IN THE SOUTH

When the line is not there

The geography

Most German wind is in the north and in the North Sea. Much of the demand is in Bavaria and Baden-Württemberg. **SuedLink**, a 700 km direct-current line between them, was due in **2022**. It is now due at the end of **2028**, at about **€10 bn**, partly because it was moved underground after local protests.

What happens meanwhile

When the wind blows, the grid operator orders northern wind farms to reduce output and southern power plants to increase it, so the line is not overloaded. This is **redispatch**. The switched-off farm is compensated; the southern plant is paid to run.

WHAT CONGESTION COSTS

Paying wind farms to stop

	COST	VOLUME	SOURCE
Germany · managing congestion (redispatch, curtailment, reserves)	**€2.95 bn (2024) · €3.08 bn (2025)**	**30.3 TWh a year**	Bundesnetzagentur, via the Bundestag
Great Britain · constraint costs, financial year 2024/25	**£1.7 bn (+64%)**	**13.5 TWh**	NESO

30 TWh is about the yearly electricity use of Ireland. It is paid for by consumers, through their bills.

NESO projects balancing costs of about £8 bn a year by 2030 unless network projects are brought forward.

BY HAND

What curtailment does to one wind farm

STEP	CALCULATION	RESULT
A 1,000 MW offshore farm at a 45% capacity factor	1,000 MW × 8,760 h × 0.45	**3,942 GWh a year**
The grid can take only 90% of it	3,942 × 10%	**394 GWh lost**
At €60 per MWh	394,200 MWh × €60	**≈ €23.7 m a year**

Whether that €23.7 m is lost depends on the contract: compensated curtailment protects the owner, uncompensated curtailment does not. Read the contract before you read the forecast.

Try it with your own numbers in the calculator on the Week 5 page.

NEGATIVE PRICES

When producers pay to produce

GERMANY	2023	2024	2025
Hours with a negative day-ahead price	**301**	**457**	**573**
Share of the year's 8,760 hours	**3.4%**	**5.2%**	**6.5%**

Sunny, windy, low-demand hours: more power than the grid can use or export. Inflexible plants and subsidised producers keep running, so the price falls below zero.

EPEX Spot (2023), SMARD / Bundesnetzagentur (2024–2025). Spain had about 555 negative hours in 2025, and 397 in the first quarter of 2026 alone.

THE CAPTURE RATE

Wind earns less than the average price

By hand · Germany, 2025

Average spot price: **8.932 ct/kWh** (€89.32/MWh).
What onshore wind actually earned on average: **7.441 ct/kWh**.
7.441 ÷ 8.932 = **83%**. That ratio is the **capture rate**.

Why

All wind farms produce at the same windy hours, so they push the price down exactly when they sell. The more wind on the system, the stronger the effect. Solar is hit harder: about **50%** in Germany in 2025.

A 2040 world with twice the wind and no new flexibility is a world with a lower capture rate.

Market values: Rödl & Partner from the German market-value data, February 2026; the ratio is computed. S&P Global gives about 82% for 2024.

INTERCONNECTION · THE TARGET IS 15% BY 2030

Can a country export its surplus?

MEMBER STATE	INTERCONNECTION, JAN 2026
Cyprus · Ireland	**0%**
Spain	**3.11%**
Poland	**5.06%**
Italy (Eni's home market)	**5.13%**
France	**5.44%**
Germany	**9.37%**

Cross-border capacity as a share of a country's generating capacity. 16 member states are above 15%; 9 are below 10%. European Commission, electricity interconnection targets.

28 APRIL 2025, 12:33

The Iberian blackout

Spain and Portugal lost power almost entirely. The final expert report (ENTSO-E, March 2026) found no single cause: oscillations, gaps in voltage control and differing voltage practices led to fast voltage increases and a cascade of power plants disconnecting.

What it does not say

It does not blame "too much solar". Be careful with that claim in your report; it is common in the press and not what the investigators found.

Why it matters here

A grid with more wind and solar needs more **control**: voltage, inertia, coordination, and links to neighbours. Spain is 3.11% interconnected.

SORTING

Turbine problem or grid problem?

1. A finished offshore farm waits a year for its export cable. **Grid**
2. Blades fail on a new turbine model and must be replaced. **Turbine**
3. Scottish farms are paid to stop because the line south is full. **Grid**
4. Wind earns 83% of the average price. **Grid: too little flexibility to move the surplus**
5. A project waits years in a connection queue. **Grid**

EXERCISE 1 · 13 MINUTES · BREAKOUT ROOMS

Where does each of your worlds hit the grid?

1. Take your team's **four worlds** from Week 4, with their 2040 wind numbers.
2. For each world, in the canvas table **"Where the grid binds"**: the **first** grid constraint it meets (connection queue, north–south lines, interconnection, negative prices, cable supply) and roughly **when**.
3. One sentence: does the world's 2040 number **survive** that constraint? If not, change the number or the world.

Back in the main room at 19:12. One person per team reads its most surprising row.

READ-OUT

Which world did the grid change?

A world that assumes 250 GW offshore by 2040 also assumes the lines, platforms and cables to land it. If your world does not say who builds them and when, it is a wish list.

This is report §4, the bottlenecks: where each one binds, and at what deployment level.

BREAK

Five minutes

Back at 19:20. Cameras on when you return.

THE GRID'S OWN SUPPLY CHAIN

Cables and transformers are scarce too

COMPONENT	TIME TO PROCURE	PRICE, REAL TERMS, SINCE 2019
Cables	**2–3 years**	**almost ×2**
Large power transformers	**up to 4 years**	**+75%**
High-voltage direct-current (HVDC) cables	**over 5 years**	**—**

Procurement times have about doubled since 2021. A handful of factories make HVDC cable, and they are booked years ahead.

IEA, 25 February 2025. Permitting is still the main cause of delay.

OFFSHORE GRIDS

The sea needs its own grid

TenneT 2 GW programme

14 standard 2 GW direct-current links in the Dutch and German North Sea, due 2031–32, running late; part of one contract was terminated in October 2025.

Danish North Sea island

Tender paused in June 2023: "too expensive and too risky".

Bornholm Energy Island

Paused in January 2025; revived in January 2026 as a 3 GW Danish–German project, with the two states **sharing** the subsidy, a first.

Offshore grids are large, shared and political: whoever pays decides whether they are built.

YOUR TWO COMPANIES

Who carries the grid risk?

Ørsted · a generator

Sells power into the grid from large offshore farms. Exposed to connection delays (Borkum Riffgrund 3), to curtailment where it is not compensated, and to falling capture prices where it sells at market prices. A long-term price contract shifts much of that risk away.

Eni · Plenitude

A smaller renewable fleet, spread across technologies and countries, and a retail business that sells power to customers. In Italy it works in a grid with 5.13% interconnection. The question for your report: does owning the customer reduce the grid risk, or only move it?

WEEK 2'S BALANCING LOOP, WITH TONIGHT'S NUMBERS

The loop that slows the transition

More wind → more congestion (+) → more curtailment and negative prices (+) → lower revenue per MWh (−) → less investment (+) → less wind (+).

One minus sign: an odd number, so it is a **balancing** loop. It does not stop growth; it slows it, and it gets stronger as wind grows.

This is the second loop your report §5 needs, next to the reinforcing learning-curve loop. Write it in the A → B (+) notation.

WHAT WEAKENS THE LOOP

Five ways out, none of them quick

Build lines

Transmission and interconnectors: the real fix, 5–15 years.

Store

Batteries move power by hours; they cut negative-price hours, not north–south distance.

Shift demand

Electric heating, vehicles and industry that use power when it is cheap.

Faster permits

RED III deadlines, the Grid Action Plan, anticipatory investment: building before the farm asks.

Price by location

Prices that are lower where power is in surplus, so new demand and plants move there. Debated, and rejected in Great Britain in 2025.

FOR YOUR FOUR WORLDS

The grid is a scenario variable

In a strong-policy world the grid is built faster and the balancing loop is weak. In a weak-policy world farms are built where the grid cannot take them, and the loop bites.

Grid build-out can even be one of your two axes, if it is not just an outcome of your policy axis. Week 4's test applies.

EXERCISE 2 · 9 MINUTES · BREAKOUT ROOMS

One grid indicator per world

1. Pick **one grid indicator** that would tell you which of your worlds is unfolding: for example negative-price hours (SMARD), constraint costs (NESO), interconnection % (European Commission), connection-queue GW (NESO, Ember).
2. Write its **baseline** with year and source, and a **value in each world** for 2030.
3. Write it in the canvas box **"One grid indicator"**. It is one of the three per world that report §7 needs.

Back at 19:49. One indicator per team, read aloud.

READ-OUT

Does your indicator separate the worlds?

If all four worlds give it the same 2030 value, it cannot tell you which world you are in, however good the source.

PHASE 1 · DUE SUNDAY 8 NOVEMBER, 23:59

Where tonight goes in the report

TONIGHT	REPORT SECTION
Where each world hits the grid; the two clocks; the queue	**§4 · The bottlenecks**
The balancing loop with its signs	**§5 · Stakeholders and the system**
Your grid indicator, with baseline and four values	**§7 · Indicators and datasets**

Sections 1–7, one voice, handed in from the report page by one of you.

CLOSE

Europe does not lack turbines. *It lacks wires.*

Every 2040 number for wind is also a statement about lines, cables, platforms and permits. Make yours say who builds them, and when.

Questions on the material: the forum of the week. Anything personal: the message panel, bottom right of every page.